

Course: Machine Learning for Robotics

Assignment# 01

Title: Shuttlecock Picking Robot

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Introduction:-

Badminton training halls and sports facilities often spend valuable time on repetitive tasks like collecting scattered shuttlecocks after drills. This assignment proposes **Shuttle Bot**, an **autonomous sports-robotics system** designed to detect shuttlecocks on the ground, navigate within a defined court-side area, pick them up safely, and place them at a **predefined collection point** with accuracy and stability.

The project combines three core areas of engineering:

- **Robotics & Mechanical Design:** building a compact, stable mobile platform with a reliable pickup/collection mechanism suitable for lightweight, irregularly shaped shuttlecocks.
- **Machine Learning-based Vision:** using a camera and an ML model (e.g., object detection) to identify shuttlecocks in real time under varying lighting and floor conditions.
- **Control & Autonomy:** integrating motion control and navigation logic so the robot can move toward detected shuttlecocks, avoid obstacles, and deliver them precisely to a target location.

The main goal is to develop a functional prototype that can perform the full workflow autonomously: **detect** → **approach** → **collect** → **transport** → **place**, while maintaining efficiency and minimizing errors such as missed detections, unstable pickup, or inaccurate placement. Beyond technical performance, the project emphasizes practical constraints such as **cost**, **ease of assembly**, **component availability**, and **expandability** (e.g., improved obstacle avoidance, better accuracy, or higher collection capacity).

To improve system reliability and real-time control performance, Shuttle Bot adopts a hybrid control architecture. The Raspberry Pi 4 is responsible for high-level processing tasks such as machine learning-based shuttlecock detection and navigation decision-making. An Arduino Uno microcontroller is integrated to handle real-time motor control, encoder feedback processing, and servo actuation. This division of tasks ensures efficient computational performance while maintaining precise motion control and system stability.

By completing this project, the team demonstrates an end-to-end application of autonomous robotics in a real sports environment, producing both a working system and a structured engineering report covering design decisions, methods, circuitry, and financial planning.

Materials and Methods:-

1. System Overview

Shuttle Bot is divided into five main subsystems:

- 1. **Mechanical Structure & Chassis**
- 2. **Drive & Motion Control System**
- 3. **Vision & Machine Learning System**
- 4. **Collection & Placement Mechanism**
- 5. **Power & Control Electronics**

Each subsystem is carefully selected to ensure stability, accuracy, and efficiency.

2. Materials (With Quantity & Specifications)

A. Mechanical Structure Components

Component	Specification	Quantity	Purpose
Aluminum Robot Chassis (2-layer)	30cm × 25cm lightweight frame	1	Base structure
Acrylic/Aluminum Mounting Plate	Custom cut	1	Mount electronics
Rubber Wheels	65mm diameter	4	Mobility
Caster Wheel	360° rotation	1	Balance support
Fasteners (Nuts, Bolts, Spacers)	M3/M4 assorted	1 set	Assembly
3D Printed Scoop/Collector	Custom design	1	Shuttle pickup
Plastic Shuttle Storage Box	15–20 capacity	1	Temporary storage

B. Drive & Motion Control System

Component	Specification	Quantity	Purpose
DC Geared Motors	12V, 200 RPM, High torque	4	Wheel drive
Motor Driver (L298N or Cytron MD10C)	Dual channel	2	Control 4 motors
Wheel Encoders	Optical type	4	Speed & distance feedback
IMU Sensor (MPU6050)	6-axis Gyro + Accelerometer	1	Orientation control

C. Vision & ML Detection System

Component	Specification	Quantity	Purpose
Raspberry Pi 4 (4GB)	Quad-core processor	1	Main ML processing unit
Raspberry Pi Camera Module v2	8MP	1	Shuttle detection
Coral USB Accelerator (Optional for high performance)	Edge TPU	1	Faster ML inference
Micro SD Card	64GB	1	OS & ML model storage

D. Collection & Placement Mechanism

Component	Specification	Quantity	Purpose
High Torque Servo Motor	20kg/cm	1	Scoop lifting
Conveyor Belt (Mini) or Roller System	6–8cm width	1	Shuttle intake
DC Motor (for conveyor)	6–12V	1	Drive intake system
Servo Motor (Placement Gate)	10kg/cm	1	Release shuttle
IR Sensor	Object detection	2	Confirm shuttle inside collector

E. Power & Control Electronics

Component	Specification	Quantity	Purpose
Li-ion Battery Pack	12V, 5000mAh	1	Main power
Buck Converter	12V to 5V	1	Power Raspberry Pi
Voltage Regulator	12V to 6V	1	Servo control
Power Switch	Heavy duty	1	Master control
Battery Charger	Compatible with Li-ion	1	Charging system
Arduino Uno	ATmega328P	1	Real-time motor & sensor control
USB Cable (Pi to Arduino)	Standard USB	1	Serial Communication
Logic Level Converter	3.3V ↔ 5V	1	Safe signal conversion

F. Obstacle Avoidance System

Component	Specification	Quantity	Purpose
Ultrasonic Sensors (HC-SR04)	2–4m range	3	Obstacle detection
IR Proximity Sensors	Short range	2	Close obstacle sensing

G. Wiring, Connectivity & Assembly Components

Component	Specification	Quantity	Purpose
Jumper Wires (Male–Male)	20cm	1 pack (40 pcs)	GPIO connections
Jumper Wires (Male–Female)	20cm	1 pack (40 pcs)	Pi to modules
Jumper Wires (Female–Female)	20cm	1 pack (40 pcs)	Module connections
Silicone Power Wires (Red/Black)	18 AWG	3 meters	Battery & motor wiring
Thin Signal Wires	22–24 AWG	3 meters	Sensor signals
Breadboard (Full size)	830 tie-points	1	Testing & prototyping
PCB / Perfboard	General purpose	1	Final soldered layout
Screw Terminal Blocks	2-pin / 3-pin	6–8	Secure motor connections
DC Power Jack Connector	Panel mount	1	Battery input
XT60 Connector (Optional)	High current	1 pair	Battery connection
Heat Shrink Tubes	Assorted sizes	1 set	Wire insulation
Electrical Tape	Standard	1 roll	Safety insulation
Cable Ties (Zip Ties)	Small	1 pack	Wire management
Mounting Standoffs	M3 Brass	1 set	Board mounting
Fuse Holder	Inline	1	Circuit protection

3. Methods (Working Mechanism)

Step 1: Shuttlecock Detection (Machine Learning)

1. Camera continuously captures frames.
2. Pre-trained object detection model (YOLOv5 or MobileNet SSD) runs on Raspberry Pi.
3. Shuttlecock bounding box coordinates are extracted.
4. Center alignment algorithm calculates turning angle and Robot aligns itself toward shuttlecock.

Step 2: Navigation & Motion Control

- PID controller regulates motor speed using encoder feedback.
- IMU ensures robot moves straight.
- Ultrasonic sensors detect obstacles.
- If obstacle detected: Robot reroutes using simple path-planning logic.

After detecting a shuttlecock, the Raspberry Pi calculates the direction and distance required. Instead of directly driving motors, it sends command signals to the Arduino via serial communication. The Arduino executes real-time PID control using encoder feedback to ensure accurate motion and stable alignment.

This hybrid structure separates computational load from time-critical motor control, improving overall performance and system robustness.

Step 3: Shuttle Collection

1. Robot stops 10–15cm before shuttlecock.
2. Servo lowers scoop.
3. Conveyor motor rotates to pull shuttle inside.
4. IR sensor confirms shuttle presence.
5. Scoop lifts back to normal position.

Step 4: Transport & Placement

1. Robot navigates to predefined drop zone using coordinate tracking.
2. Stops at marked location.
3. Servo opens release gate.
4. Shuttlecock drops precisely inside collection box.

4. Software Architecture

Programming Language:

- Python (Raspberry Pi – ML & Navigation)
- C++ (Arduino – Motor Control & PID)

Libraries Used:

- OpenCV
- TensorFlow Lite / PyTorch
- PySerial (Pi-Arduino Communication)
- Arduino PID Library
- Servo Library

5. Innovation & Cutting-Edge Features

To achieve maximum marks in design:

- Real-time object detection using YOLO.
- Closed-loop motor control with encoders.
- Adaptive lighting correction in vision system.
- Modular mechanical intake system.
- Energy-efficient power management.

6. System Workflow Summary

Camera → ML Detection → Angle Calculation → PID Navigation → Shuttle Intake → Obstacle Avoidance → Drop Zone Navigation → Precise Placement

Circuit Diagram:-

1. System Architecture Overview

The Shuttle Bot uses a distributed control system:

- Raspberry Pi 4
 - Runs object detection model (YOLO / TensorFlow Lite)
 - Processes camera input
 - Calculates navigation path
 - Sends movement commands via Serial communication
- Arduino Uno
 - Controls motor drivers (L298N)
 - Reads encoder feedback
 - Executes PID motor speed control
 - Operates servo motors
 - Reads ultrasonic and IR sensors

Communication between Raspberry Pi and Arduino is achieved through USB Serial protocol. The Raspberry Pi sends high-level commands such as MOVE_FORWARD, TURN_LEFT, STOP, or COLLECT, while Arduino executes precise hardware control operations.

2. Power Distribution Circuit

Main Power Source

- 12V 5000mAh Li-ion Battery

Power Flow:

12V Battery →

→ Direct to Motor Drivers (12V input)

→ Buck Converter (12V to 5V) → Raspberry Pi + Camera

→ Voltage Regulator (12V to 6V) → Servo Motors

Wiring Details

Connection	From	To	Notes
Battery +	Battery	Power Switch	Master control
Switch Output	Switch	Distribution Bus	Main supply line
12V Line	Bus	Motor Driver VIN	Motor power
12V Line	Bus	Buck Converter IN	5V generation
12V Line	Bus	6V Regulator IN	Servo supply
5V Output	Buck Converter	Raspberry Pi 5V	Stable logic voltage
6V Output	Regulator	Servo Motors	High torque operation
GND	All components	Common Ground	Must be shared

❖ Important: All grounds must be connected together to avoid unstable signals.

3. Motor Driver Circuit (4 DC Motors)

We use **2 Dual-Channel Motor Drivers (L298N)**.

Each driver controls 2 motors.

Motor Connections

Motor	Driver Channel
Front Left	Driver 1 OUT1 & OUT2
Rear Left	Driver 1 OUT3 & OUT4
Front Right	Driver 2 OUT1 & OUT2
Rear Right	Driver 2 OUT3 & OUT4

Control Pins (Example GPIO Mapping)

Driver Pin	Raspberry Pi GPIO
IN1	GPIO 17
IN2	GPIO 27
IN3	GPIO 22
IN4	GPIO 23
ENA	GPIO 18 (PWM)
ENB	GPIO 19 (PWM)

Same pattern repeated for second driver.

PWM pins control speed.

IN pins control direction.

4. Encoder Wiring

Each motor has an encoder.

Encoder Pin	Raspberry Pi GPIO
Encoder A (FL)	GPIO 5
Encoder B (FL)	GPIO 6
Encoder A (FR)	GPIO 12
Encoder B (FR)	GPIO 13
Encoder A (RL)	GPIO 16
Encoder B (RL)	GPIO 20
Encoder A (RR)	GPIO 21
Encoder B (RR)	GPIO 26

Encoders provide pulse feedback for:

- Distance measurement
- Speed regulation
- PID control

5. Camera Connection

Raspberry Pi Camera Module connects via:

CSI Ribbon Cable → Raspberry Pi CSI Port

No external wiring required.

Powered directly from Raspberry Pi.

6. Ultrasonic Sensor Circuit (3 Units)

Each ultrasonic sensor has 4 pins:

- VCC (5V)
- GND
- TRIG
- ECHO

Example GPIO Mapping:

Sensor	TRIG	ECHO
Front	GPIO 24	GPIO 25
Left	GPIO 8	GPIO 7
Right	GPIO 1	GPIO 0

- ❖ Use voltage divider for ECHO pin (because Raspberry Pi works at 3.3V logic).

7. IR Shuttle Detection Sensors

Sensor	GPIO
Collector IR 1	GPIO 9
Collector IR 2	GPIO 10

Used to confirm shuttle presence inside storage.

8. Servo Motor Connections

High Torque Servo (Scoop):

- Signal → GPIO 11 (PWM)
- VCC → 6V Regulated Output
- GND → Common Ground

Release Gate Servo:

- Signal → GPIO 14 (PWM)
- VCC → 6V Regulated Output
- GND → Common Ground

9. Conveyor Motor Circuit

Conveyor DC Motor connected to:

- Small motor driver channel
- Controlled via Raspberry Pi GPIO (PWM + Direction)

10. Communication & Data Flow

Camera → Raspberry Pi → ML Detection → Command Generation →
Serial Communication → Arduino → Motor Drivers → Motors

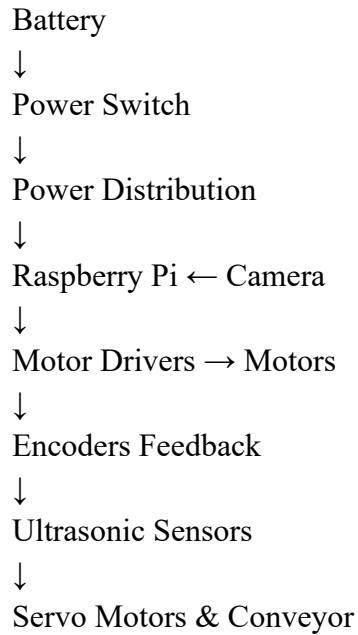
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Encoder Feedback → Arduino PID Correction

11. Safety Features

- Fuse between battery and main circuit
- Heat sinks on motor drivers
- Proper cable management
- Emergency manual kill switch

12. Simplified Block Diagram (Logical View)



13. Hybrid Control Wiring Description

The Raspberry Pi and Arduino Uno are connected using USB serial communication. The Arduino is powered through the 12V battery via Vin or external regulated supply.

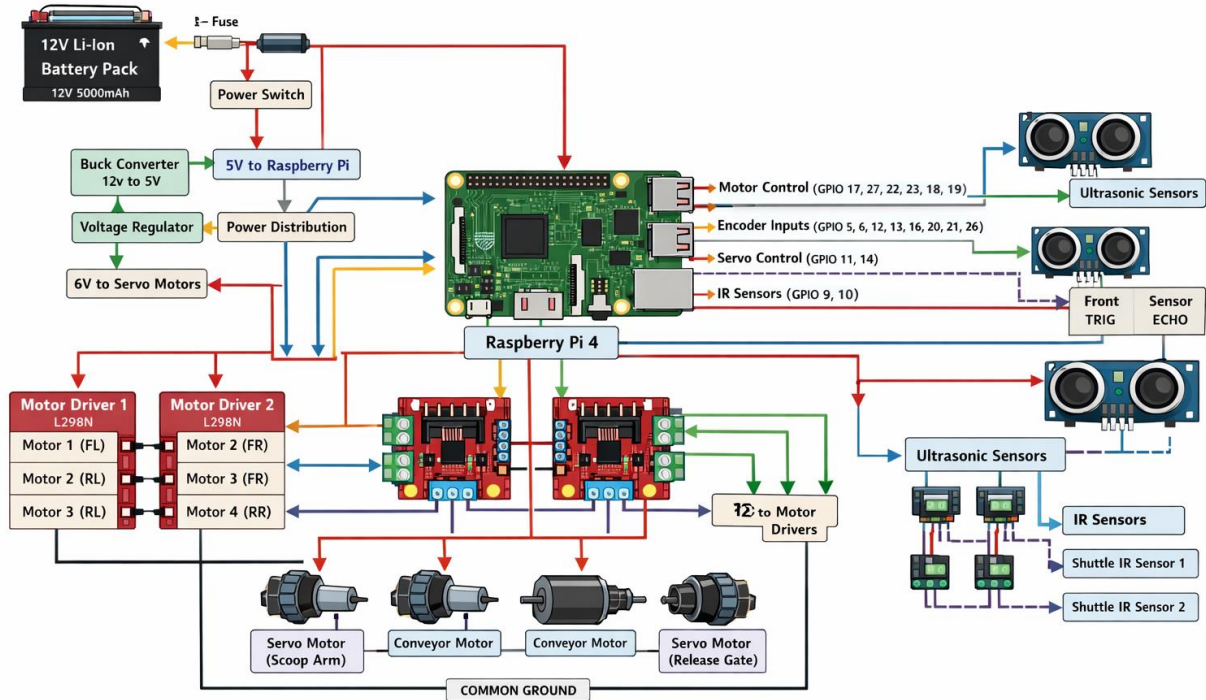
Connections:

- Raspberry Pi USB → Arduino USB (Serial Communication)
- Arduino Digital Pins → Motor Driver IN1–IN4
- Arduino PWM Pins → ENA / ENB
- Arduino Pins → Ultrasonic TRIG & ECHO
- Arduino Pins → Servo Signal
- Common Ground between Raspberry Pi, Arduino, Motor Drivers, and Power Supply

The Raspberry Pi does not directly control motors. It acts as the AI brain, while Arduino acts as the motion controller.

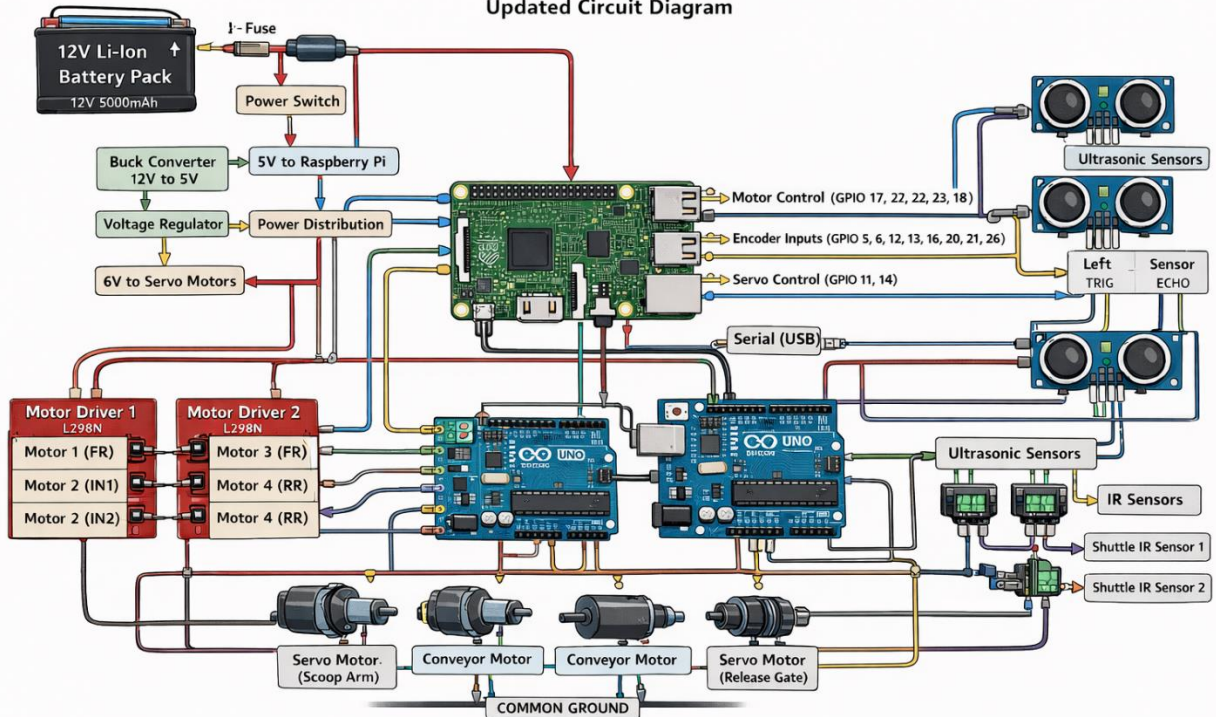
Shuttle Bot

Autonomous Shuttlecock Collector System



Shuttle Bot

Updated Circuit Diagram



Financial Analysis:-

The financial analysis breaks down the cost of all major components, additional expenses, and the **total budget** required to complete the Shuttle Bot project.

1. Mechanical & Structural Components

Component	Qty	Estimated Price (PKR)
Aluminum Robot Chassis	1	1,500
Rubber Wheels (65mm)	4	1,200
Caster Wheel	1	300
Fastener Set (nuts/bolts/spacers)	1	500
3D Printed Collection Scoop	1	800
Shuttle Storage Box	1	500
Subtotal Mechanical: PKR 4,800		

2. Drive & Motion Control System

Component	Qty	Estimated Price (PKR)
DC Geared Motors (12V, high torque)	4	4,000
Motor Drivers (L298N)	2	1,000
Wheel Encoders	4	1,600
IMU (MPU6050)	1	700
Subtotal Drive Control: PKR 7,300		

3. Vision & Processing System

Component	Qty	Estimated Price (PKR)
Raspberry Pi 4 (4GB)	1	12,000
Raspberry Pi Camera Module	1	3,000
Coral USB Accelerator (Optional)	1	5,000
Micro SD Card (64GB)	1	1,500

Subtotal Vision & Processing: PKR 21,500

Note: If you'll not use the Coral Accelerator, remove PKR 5,000 New subtotal = PKR 16,500.

4. Control & Logic Components

Component	Qty	Estimated Price (PKR)
Arduino Uno	1	1,300
USB Cable (Pi-Arduino)	1	300
Logic Level Converter (3.3V ↔ 5V)	1	500

Subtotal Control Components: PKR 2,100

5. Power System

Component	Qty	Estimated Price (PKR)
12V Li-ion Battery Pack (5000mAh)	1	4,000
Buck Converter (12V → 5V)	1	600
Voltage Regulator (12V → 6V)	1	500
Battery Charger	1	1,200
Power Switch + Fuse	1	300

Subtotal Power: PKR 6,600

6. Sensors & Mechanism Components

Component	Qty	Estimated Price (PKR)
Ultrasonic Sensors (HC-SR04)	3	900
IR Sensors	2	300
Servo Motors (High Torque)	2	2,000
Conveyor DC Motor	1	800
Subtotal Sensors & Mechanism: PKR 4,000		

7. Miscellaneous & Contingencies

Expense	Estimated Price (PKR)
Wires / Connectors / Headers	1500
Heat sinks / Mounting hardware	500
Shipping / Miscellaneous	800
Subtotal Miscellaneous: PKR 2,800	

Total Estimated Cost

Category	Cost (PKR)
Mechanical & Structure	4,800
Drive & Motion Control	7,300
Vision & Processing	21,500
Control & Logic	2,100
Power System	6,600
Sensors & Mechanisms	4,000
Miscellaneous	2,800

Total Project Cost (Approx.):

PKR 49,100

(If Coral USB Accelerator not used → PKR 44,100)

Teamwork & Project Management:-

1. Team Structure

The Shuttle Bot project was developed through structured collaboration among a team of 3–4 members. Each member was assigned a clearly defined technical role to ensure efficiency, accountability, and smooth progress.

Team Roles:

1. Team Captain / Project Lead

- Managed project timeline and milestones
- Coordinated communication between members
- Oversaw integration of hardware and software
- Ensured deliverables met assignment requirements

2. Hardware & Mechanical Engineer

- Designed robot chassis structure
- Installed motors, wheels, and collection mechanism
- Managed assembly and wiring layout
- Ensured mechanical stability

3. Electronics & Control Engineer

- Designed circuit architecture
- Connected Raspberry Pi and Arduino system
- Implemented motor drivers and sensor wiring
- Tested encoder and PID control system

4. Software & Machine Learning Engineer

- Developed shuttlecock detection model
- Integrated OpenCV and TensorFlow Lite
- Implemented navigation logic
- Managed Raspberry Pi–Arduino serial communication

2. Collaboration Strategy

The team adopted a structured workflow divided into stages:

Phase 1: Planning & Research

- Analyzed project requirements
- Selected hybrid architecture (Raspberry Pi + Arduino)
- Conducted component comparison

Phase 2: Hardware Development

- Assembled chassis
- Installed motors and wiring
- Mounted camera and sensors

Phase 3: Software Development

- Trained or implemented object detection model
- Developed Arduino PID motor control
- Implemented serial communication protocol

Phase 4: Integration & Testing

- Combined hardware and software systems
- Conducted trial runs
- Debugged navigation and pickup issues
- Optimized performance

3. Leadership & Communication

The team captain ensured:

- Weekly progress meetings
- Task allocation based on expertise
- Clear documentation of system updates
- Conflict resolution through structured discussion

Communication tools included:

- WhatsApp group for quick updates
- Shared Google Drive for documentation
- GitHub (optional) for code version control

4. Challenges & Solutions

Challenge	Solution
ML processing delay	Optimized image resolution
Motor drift issue	Implemented PID control using encoders
Power instability	Added voltage regulation & common grounding
Sensor noise	Applied filtering and threshold tuning

5. Learning Outcomes

Through this project, the team gained practical experience in:

- Autonomous robotics design
- Machine learning implementation in embedded systems
- Hybrid control architecture (Microcontroller + SBC)
- Real-world debugging and system integration
- Cost planning and engineering documentation

6. Conclusion of Project Execution

The Shuttle Bot project successfully demonstrates the integration of machine learning, control systems, and robotics engineering into a functional autonomous sports assistant. The hybrid Raspberry Pi and Arduino architecture enhanced system performance, reliability, and modularity.

This structured teamwork approach ensured the project objectives were achieved efficiently while maintaining innovation and technical depth.

Conclusion:-

The Shuttle Bot project successfully demonstrates the integration of machine learning, robotics, and control systems into a practical autonomous sports solution. By combining a Raspberry Pi 4 for high-level vision processing and an Arduino Uno for real-time motor control, the system achieves both computational efficiency and precise motion execution.

The implemented hybrid architecture ensures reliable shuttlecock detection, stable navigation using PID control, and accurate pickup and placement mechanisms. The inclusion of encoder feedback, obstacle avoidance sensors, and regulated power distribution enhances system robustness and operational safety.

From a technical perspective, the project highlights the importance of separating computational tasks from time-critical control operations. From a practical perspective, it presents a scalable and cost-effective automation solution for sports training environments.

Future improvements may include advanced path planning algorithms, higher-frame-rate object detection using AI accelerators, SLAM-based localization, and increased collection capacity. Overall, Shuttle Bot represents a strong application of autonomous robotics in real-world environments and fulfills the objectives outlined in the assignment.

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